ASSIGNMENT-2

Foundations of Cryptography

Course: BCA | Subject: Cryptography

Q1. List the four fundamental cryptographic goals/objectives.

The four fundamental cryptographic objectives are confidentiality, integrity, authentication, and non-repudiation.

Q2. Define confidentiality as a cryptographic goal with one example.

Confidentiality means that information can be accessed only by authorized people. For example, encryption protects online banking details so that a person intercepting the network traffic cannot read them.

Q3. Explain confidentiality, integrity, authentication and non-repudiation with real-world scenarios. Why are all four important together?

Confidentiality protects information from being read by unauthorized people. For example, encryption keeps a private message unreadable to an eavesdropper. Integrity ensures that data has not been altered without detection; a message authentication code can help detect a changed message. Authentication verifies the identity of a user or the source of a message; for example, a website certificate helps a browser verify the website. Non-repudiation provides evidence that a particular party performed an action, often through a digital signature. All four are important because a message might be private but altered, unchanged but sent by an impostor, or signed using a key whose owner cannot be reliably established. Together, suitable mechanisms make communication more secure and trustworthy.

Q4. What is the difference between a substitution cipher and a transposition cipher? Give one example of each.

A substitution cipher replaces letters or groups of letters with other symbols; the Caesar cipher is an example. A transposition cipher keeps the same letters but changes their positions; the Rail Fence cipher is an example.

Q5. Discuss the evolution from classical ciphers to modern standards (DES → AES, RSA, SHA).

Classical ciphers such as Caesar and Vigenère used substitution and transposition methods. They were often vulnerable to language patterns and frequency analysis. Mechanical systems such as Enigma made encryption more complex, but cryptanalysis during World War II demonstrated the importance of mathematics and operational security. DES was adopted as a standard in 1977 and became an important computer-based symmetric cipher, but its effective 56-bit key eventually became too short against modern brute-force attacks. AES was selected as its successor and standardized in 2001, providing a 128-bit block and 128-, 192- or 256-bit keys. RSA introduced public-key cryptography into practical use, supporting public/private key operations such as digital signatures and key establishment. SHA is a family of hash functions used for integrity checks and as part of signature systems; hashing is not encryption. These changes were driven by increasing computing power, the need for stronger keys, internet communication and scalable key management.

Q6. What is a key in cryptography, and why is key management important?

A cryptographic key is a value used by an algorithm to encrypt, decrypt, sign or verify data. Key management is important because keys must be generated, distributed, stored, rotated and revoked securely; if a key is exposed, even a strong algorithm may no longer protect the information.

Q7. Explain how cryptography protects data during transmission, prevents unauthorized access and detects tampering in networks.

Cryptographic protocols such as TLS encrypt data during transmission to protect confidentiality. Certificates and authentication methods help verify the identity of servers and, when configured, users. VPN technologies such as IPsec protect traffic between devices or networks. Message authentication codes and authenticated encryption help detect unauthorized changes to data. Digital signatures can verify that a message or software update was signed by the holder of a private key. Passwords should be stored using a suitable salted password-hashing scheme rather than reversible encryption. These mechanisms help defend against eavesdropping, impersonation, unauthorized access and tampering.

Q8. Differentiate between a passive attack and an active attack, with one example of each.

A passive attack observes communication without changing it, such as eavesdropping on unencrypted network traffic. An active attack modifies, injects or disrupts communication, such as changing a message in transit or replaying a captured authentication message.

Q9. Describe the evolution from symmetric-only systems to asymmetric cryptography and how it addressed key distribution.

Traditional symmetric systems use the same secret key for encryption and decryption. They are efficient, but communicating parties must share the key securely in advance; this becomes difficult when many users need to communicate. Public-key cryptography, publicly introduced by Diffie and Hellman in 1976, uses a public key and a related private key. The public key can be shared openly, while the private key is kept secret. This enables methods for establishing shared secrets and creating or verifying digital signatures without sending the private key over the network. RSA followed in 1977. Modern systems commonly combine approaches: asymmetric techniques authenticate parties or establish a session key, and fast symmetric encryption protects the actual data. This evolution responded to growing networks, more powerful computers and the need for scalable secure communication.

Q10. Give one example each of a modern symmetric algorithm and a modern asymmetric algorithm.

A modern symmetric algorithm is AES, which uses a shared secret key. A modern asymmetric algorithm is RSA, which uses a public/private key pair. Elliptic-curve cryptography is another widely used asymmetric approach.

Q11. Discuss Classical Cryptography and Modern Cryptography and the transition to computer-based algorithms.

Classical cryptography includes manual substitution and transposition methods such as Caesar, Vigenère and Rail Fence ciphers. These methods often preserved language patterns and could be attacked using frequency analysis. Mechanical systems such as Enigma automated more complicated transformations and were used during World War II. Modern cryptography uses mathematical algorithms designed for computers, including symmetric ciphers such as DES and AES, public-key systems such as RSA and elliptic-curve cryptography, and hash functions such as SHA-256. Key milestones include Shannon's theoretical work, public-key cryptography in the 1970s, DES standardization in 1977 and AES standardization in 2001. Protocols such as TLS and IPsec apply cryptography to modern networks. This transition was driven by the need to protect large volumes of digital information, manage keys across many users and withstand more powerful computational attacks.

References

1. William Stallings, Cryptography and Network Security: Principles and Practice.

2. NIST publications on DES, AES and Secure Hash Standards.

3. Whitfield Diffie and Martin E. Hellman, New Directions in Cryptography, 1976.